



Digital Filter 1

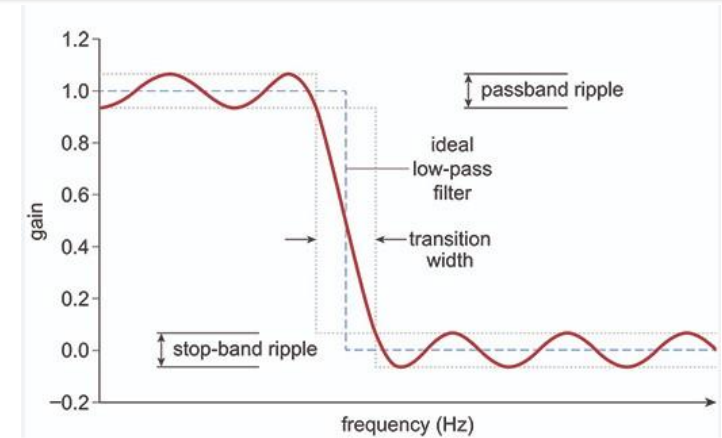
デジタルフィルタ1

デジタルフィルタの基礎概念と FIR フィルタについて理解しよう

Lets understand the basic concepts of digital filters and FIR filters.

教科書の該当範囲：第6章.

Textbook Coverage: Chapter 6



フィルタ "filter" の入出力特性をどのように表現するか？

How to express the input / output characteristics of a filter

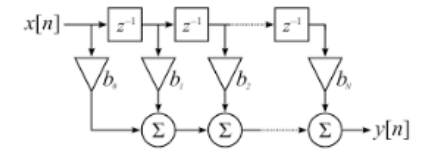
周波数応答 "spectral response": 周波数スペクトル的な入出力の関係で表現

Spectral response : Expressed as the relationship between input and output in terms of frequency spectrum.

伝達関数 "transfer function": (連続周波数版, 離散周波数版)

Transfer function: (Continuous frequency version, discrete frequency version)

TF = output signal / input signal → TF is constructed using Laplace Transform



ブロック図：入カスペクトル $X(\omega) \rightarrow$ フィルタ $H(\omega) \rightarrow$ 出カスペクトル $Y(\omega) = H(\omega)X(\omega)$

Block diagram : input spectrum $X(\omega) \rightarrow$ filter $H(\omega) \rightarrow$ Output spectrum $Y(\omega) = H(\omega)X(\omega)$

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} \quad , \quad H_k = \frac{Y_k}{X_k}$$

アナログフィルタの取扱の際に好都合

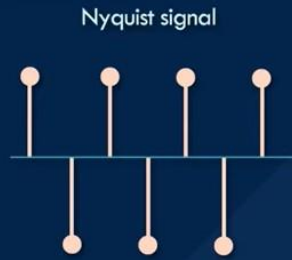
Convenient when working with analog filters

ディジタルフィルタ(DFT フィルタ)の実装にも利用可能だが、区間毎のバッファ処理となり、レイテンシ “latency” (処理遅延) が問題

It can also be used to implement digital filters (DFT filters) but because buffer processing is performed for each section, latency(processing delay) is an issue



Test signals

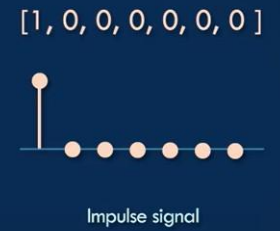
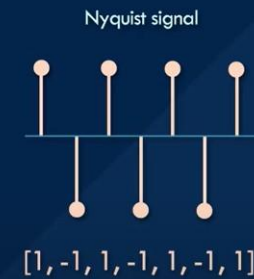
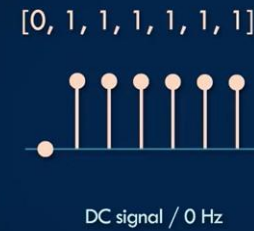


Algorithmic building blocks

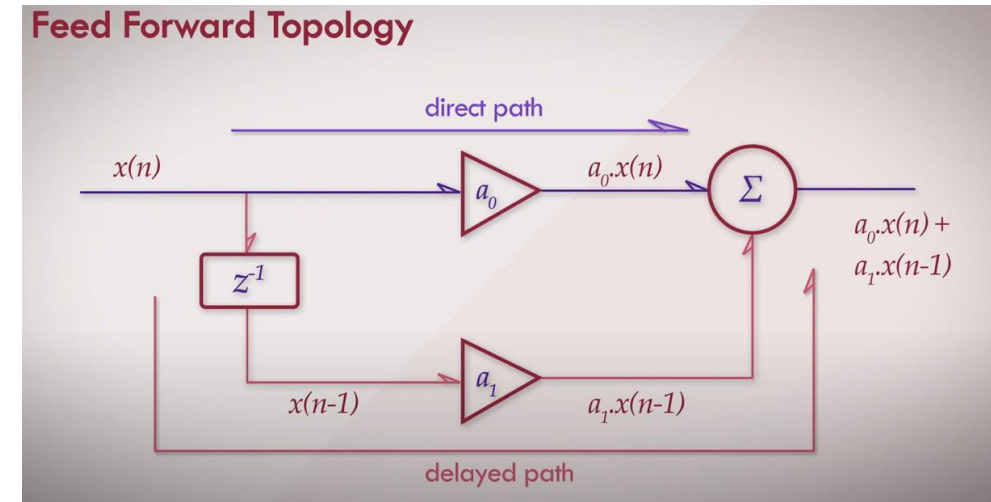


Test signals

simple, concise and perhaps periodic



Feed Forward Topology



伝達関数の具体的な数式の定義は容易

It is easy to define the specific formula for the transfer function

周波数(単独の数値)を引数とした普通の関数. 適当(LPFとかBEFとか所望の特性を表すよう)な数式を定義すれば OK

It is a normal function with frequency (a single numerical value) as an argument. Just define an appropriate formula (such as LPF or BEF, which represents the desired characteristics) and you're good to go.

時間応答 "temporal response": 時間波形的な入出力の関係で表現

Temporal response: Expressed as a time waveform relationship between input and output.

応答関数 "response"

$$h(x(t)) = \dots \quad \text{orz}$$

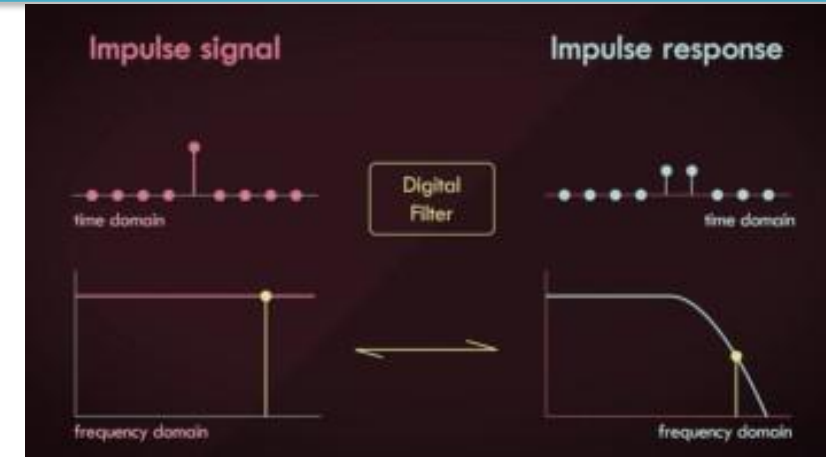
ブロック図: 入力波形 $x(t)$ → ノイлта $h(x)$ → 出力波形 $y(t)=h(x(t))$

Block diagram: Input waveform $x(t)$ → filter $h(x)$ → Output waveform $y(t)=h(x(t))$

応答関数をどのように定義すれば? How do we define the response function?

波形(関数, 数値列)を引数とする特殊な関数(汎関数と云われるもの)ですよ. → インパルス応答と Z 変換を利用

It is a special function (**called a functional**) that takes a waveform (function, numeric sequence) as an argument. → Use impulse response and **Z transform**



インパルス “impulse” (一瞬だけのパルス) 入力に対するフィルタ出力

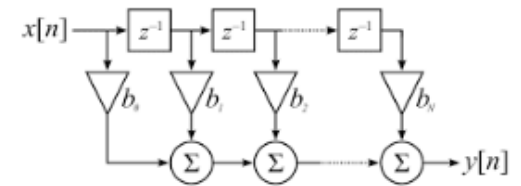
Filter output for impulse input.

複数のインパルスの遅延・定数倍・総和で任意の波形を表現可能.

Arbitrary waveforms can be expressed by **delaying**, **multiplying by a constant**, and **adding up multiple impulses**.

インパルスをズラしながら大きさを調整して重ね合わせるってことね

It involves **shifting the impulses**, **adjusting their size**, and **overlapping them**



インパルス応答と入力波形を畳み込めば、フィルタの時間応答になる

Convolving the impulse response with the input waveform results in the time response of the filter.

→ 両者をズラしながら乗算し、合計するのが畳み込み "convolution"

Convolution involves multiplying the two with a shift and adding them together.

→ なお、フィルタの周波数応答は、伝達関数と入力スペクトルの積でした。畳み込みとフーリエ変換の積とは表裏一体

The **frequency response** of a filter is the **product of the transfer function** and the **input spectrum**. Convolution and the product of the Fourier transform are two sides of the same coin.

This computation is also known as Convolution →

$$y[n] = b_0 x[n] + b_1 x[n-1] + \dots + b_N x[n-N]$$

$$= \sum_{i=0}^N b_i \cdot x[n-i],$$

インパルスのフーリエ変換は一定値

The Fourier transform of an impulse is a constant value

→ スペクトル分布に、あらゆる周波数が含まれる

The spectral distribution includes all frequencies

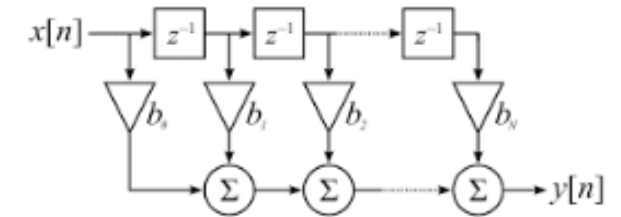
インパルス応答のフーリエ変換は伝達関数そのもの

The Fourier transform of the impulse response is the transfer function itself

→ てことは、伝達関数の逆フーリエ変換はインパルス応答ですね

This means that the inverse Fourier transform of the transfer function is the impulse response

I FT of Transfer function = Impulse Response



バリエーション: Variations

Output response of FIR $Y[n] \rightarrow$ depends only on the input signal

有限インパルス応答 (FIR) "finite impulse response": 単独のインパルス入力に対して, 有限個のパルスを出力

Finite impulse response (FIR): A finite number of pulses are output in response to a single impulse input

→例えば, 短めの残響 "reverb" みたいなものですよーお

For example, it's like a short reverberation "reverb"

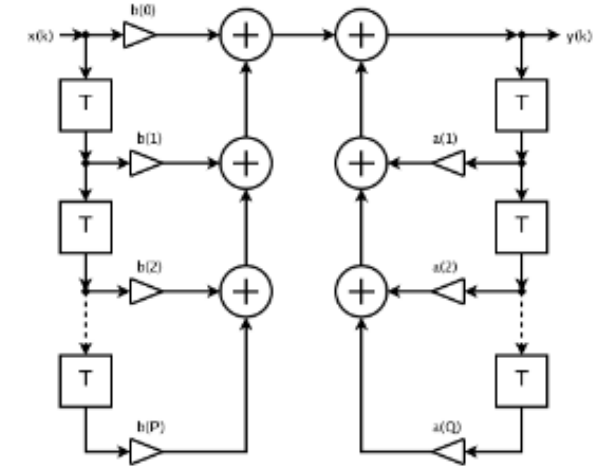
$$y[n] = b_0 x[n] + b_1 x[n-1] + \dots + b_N x[n-N]$$

$$= \sum_{i=0}^N b_i \cdot x[n-i],$$

$X[n]$ – input signal, $y[n]$ – output signal, N filter order, N th – order filter has $N + 1$ terms
 b_i – value of impulse response at the i th instant for $0 \leq i \leq N$ of an N th order FIR filter.

バリエーション: Variations

Output response of IIR $Y[n] \rightarrow$ Influenced by Input signal $X[n]$ + feedback from previous output



無限インパルス応答 (IIR) "infinite impulse response" 単独のインパルス入力に対して, 無限個のパルスを出力

Infinite impulse response (IIR) An infinite number of pulses are output for a single impulse input

→ 長めの残響ですかね

Long reverberation

$$\text{IIR Filter Equation: } y(n) = \sum_{k=0}^N a(k)x(n-k) + \sum_{j=0}^P b(j)y(n-j)$$

Output used recursively

具体例を元にして, FIR の時間応答を導出してみる

Lets use a specific example to derive the time response of FIR

インパルス応答の例: (残響時間 $J = 3$ の場合)

Example of an impulse response (for reverberation time $J = 3$):

$$\mathbf{x} = \begin{pmatrix} x_0 \\ 0 \\ 0 \\ 0 \\ \vdots \end{pmatrix} \rightarrow \mathbf{y} = \begin{pmatrix} b_0 x_0 \\ b_1 x_0 \\ b_2 x_0 \\ 0 \\ \vdots \end{pmatrix}$$

時間 $\sim n = 0 \sim (J-1)$ に対して, 出力は $y_n = b_n x_0$ ですね

For time $\sim n = 0 \sim (J-1)$, the output is $y_n = b_n x_0$

遅延インパルス応答の例：（同上かつ遅延時間 $m=2$ の場合

Example of delayed impulse response: (same as above with delay time $m=2$)

$$\mathbf{x} = \begin{pmatrix} 0 \\ 0 \\ x_2 \\ 0 \\ 0 \\ 0 \\ \vdots \end{pmatrix} \rightarrow \mathbf{y} = \begin{pmatrix} 0 \\ 0 \\ b_0 x_2 \\ b_1 x_2 \\ b_2 x_2 \\ 0 \\ \vdots \end{pmatrix}$$

時間 $n = m \sim (m + J - 1)$ に対して,出力は $y_n = b_n - m^x m$ ということ

For time $n = m \sim (m + J - 1)$ the output is $y_n = b_n - m^x m$

あるパルス波形の時間応答の例：（残響時間 $J=3$ の場合）

Example of time response of a pulse waveform: (for reverberation time $J=3$)

インパルス応答の重ね合わせの単純な例でした

This was a simple example of superposition of impulse responses.

$$\mathbf{x} = \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \\ 0 \\ 0 \\ 0 \\ \vdots \end{pmatrix} \rightarrow \mathbf{y} = \begin{pmatrix} b_0 x_0 \\ b_1 x_0 + b_0 x_1 \\ b_2 x_0 + b_1 x_1 + b_0 x_2 \\ b_2 x_1 + b_1 x_2 + b_0 x_3 \\ b_2 x_2 + b_1 x_3 \\ b_2 x_3 \\ 0 \\ \vdots \end{pmatrix}$$

任意波形に対する時間応答

Time response to arbitrary waveforms

$$y_n = \sum_m b_m x_{n-m}$$

現在の出力 y_n に対して、過去の入力 x_{n-m} が比率 b_m だけ配分されている。これが、入力の波形 x とフィルタのインパルス応答 b との畳み込みです

For the current output y_n the past input x_{n-m} is allocated by a ratio b_m . This is the convolution of the input waveform x with the filter impulse response b .

残響の標本数 J の場合、総和の遅延番号 m の範囲は $0 \sim (J-1)$ です。

プログラミング的には、 xn とかの添字付き変数は配列要素 $x[n]$ とかですよ

For the number of reverberation samples J , the sum delay number m is in the range $0 \sim (J-1)$. In programming terms, subscripted variables such as xn are array elements such as $x[n]$.

Z 変換 (ZT) “Z transform” は DFT の変種 The Z transform (ZT) is a variant of the DFT

→ DFT:

$$X_k = \sum_n x_n W_N^{kn} = \sum_n x_n \exp\left(-j \frac{2\pi kn}{N}\right)$$

複素変数 z の導入 Introduction of complex variables

$$z = W_N^{-k} = \exp\left(j \frac{2\pi k}{N}\right)$$

→ この z は、本来は任意の複素数なのですが.. とりあえず、回転子 WN の更なる短縮記法と思えばヨシ

This z is actually an arbitrary complex number, but for now, you can think of it as a further abbreviation for the rotator WN



ZT :

$$X(z) = \sum_n x_n z^{-n}$$

本来は、関数の多項式展開（テイラー展開とか）の複素数版なんですが... とりあえず、シンプルな数式だなーと思えばヨシ

Originally, it is a complex version of polynomial expansion of a function (such as Taylor expansion), but for now, it's fine if you think it's a simple formula

謎の変数 z を使っていますが、周波数 ω とか f とか k の関数と思えば、この $X(z)$ もスペクトルですよ

Although we use the mysterious variable z , if we think of it as a function of frequency ω , f , or k , then this $X(z)$ is also a spectrum

総和の標本番号 n の範囲は $0 \sim (N-1)$ です. また, N は標本数でした

The range of the sample number n of the summation is $\sim 0 \sim (N-1)$. Also, N is the number of samples

時間応答を元にして, FIR の周波数応答を導出してみる

Based on the time response, let's derive the frequency response of an FIR filter

時間応答 : $y_n = \sum_m b_m x_{n-m}$

Time Response

Z 変換 :

Z Transform

$$\begin{aligned} Y(z) &= \sum_n \left(\sum_m b_m x_{n-m} \right) z^{-n} \\ &= \sum_n \sum_m \left(b_m x_{n-m} z^{-(n-m)} z^{-m} \right) \\ &= \sum_m \left(b_m z^{-m} \sum_n x_{n-m} z^{-(n-m)} \right) \end{aligned}$$

ここで, $\sum_n$ は全標本値に対する総和だよね? そして, x_{n-m} と $z^{-(n-m)}$ は周期的なので, 添字 $n-m$ をズラし, n にしても, 総和の値は不変なので無問題

Here, $\sum_n$ is the sum over all sample values, right? And since x_{n-m} and $z^{-(n-m)}$ are periodic, even if we shift the subscript $n-m$ to n , the value of the sum remains unchanged, so there is no problem.

$$\begin{aligned} Y(z) &= \left(\sum_m b_m z^{-m} \right) \left(\sum_n x_n z^{-n} \right) \\ &= H(z) X(z) \end{aligned}$$

伝達関数 (z 領域)

Transfer Function (z Domain)

$$H(z) = \sum_m b_m z^{-m}$$

z^{-m} : 入力に対する時間 m の遅延子 "**delay operator**"

z^{-m} : A **delay operator** of time m for the input

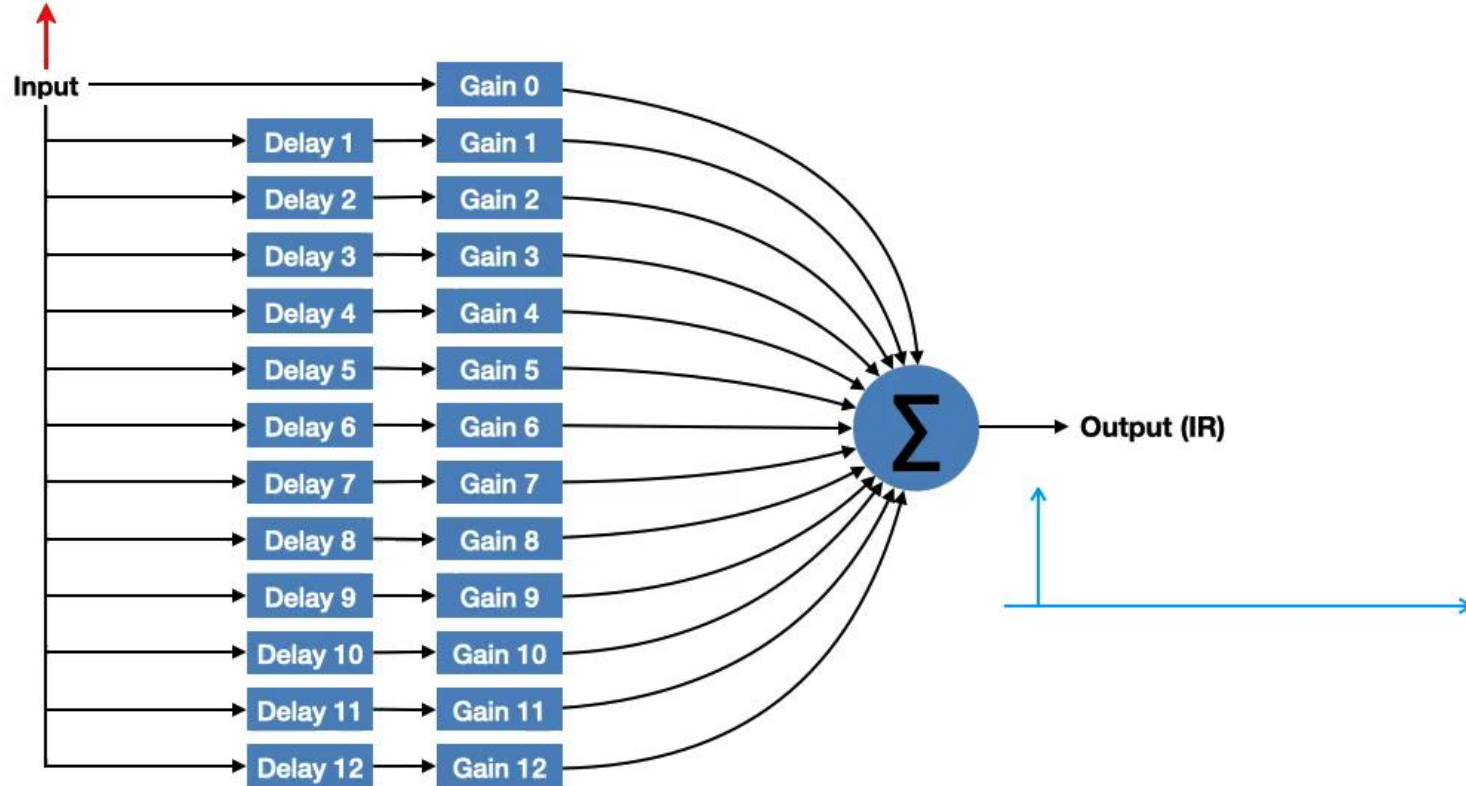
or 遅延演算子. 単なる複素数に見えますが, 回転子と同様に, 演算子と見なされます

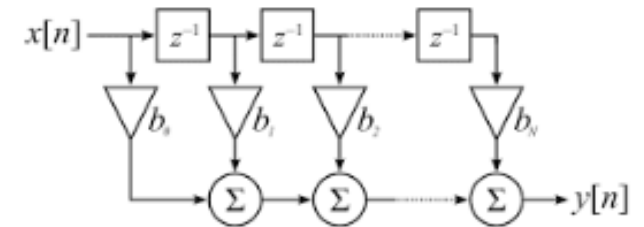
or Delay operator. It looks like a simple complex number, but like the rotator, it is considered an operator



FIR フィルタの周波数応答

Frequency response of an FIR filter





b_m : 遅延された入力 x_{n-m} に対するフィルタ係数 “filter coefficient”
 b_m : Filter coefficient for the delayed input x_{n-m}

伝達関数と云いつつ z 領域では、周波数応答だけでなく、時間応答も表現している？

Although it is called a transfer function, in the z domain, does it express not only the frequency response but also the time response

伝達関数の図式表現: ブロック図 “block diagram”

Diagrammatic representation of transfer function: Block diagram “block diagram”

$$\begin{aligned} Y(z) &= \left(\sum_m b_m z^{-m} \right) \left(\sum_n x_n z^{-n} \right) \\ &= H(z) X(z) \end{aligned}$$

$$H(z) = \sum_m b_m z^{-m}$$

遅延素子 "delay element": z^{-1} は、現在の入力 x_n に対して、過去の入力 x_{n-1} を出力

The delay element z^{-1} outputs the past input x_{n-1} for the current input x_n

メモリみたいなものですね。過去の入力を1個しか保存できませんが

It's like a memory. It can only store one past input

乗算器 "multiplier": 入力を b_m 倍して出力

Multiplier: Multiplies the input by b_m and outputs it

加算器 "adder": 複数の入力を合計し出力

Adder: Adds multiple inputs and outputs them

ブロック図では、Z 変換のフィルタと時間領域の信号とが混在

The block diagram shows a mixture of Z-transform filters and time-domain signals

$$\begin{aligned} Y(z) &= \left(\sum_m b_m z^{-m} \right) \left(\sum_n x_n z^{-n} \right) \\ &= H(z)X(z) \end{aligned}$$

$$H(z) = \sum_m b_m z^{-m}$$

伝達関数(周波数領域): $H_k = \sum_m b_m \exp\left(-j\frac{2\pi km}{N}\right)$

Transfer Function (Frequency Domain)

FIR の場合, 伝達関数の変数変換 ($Z \rightarrow$ 周波数領域) は単純に, z を $\exp \cdots$ へ置き換えるだけで OK. 次回の IIR の場合, 単純ではない

In the case of FIR, the variable transformation of the transfer function ($Z \rightarrow$ frequency domain) is simply to replace z with $\exp \dots$
In the case of IIR, it is not so simple

$$y_n = (x_n + x_{n-1})/2$$

$$H(z) = (1 + z^{-1})/2$$

$$\begin{aligned} |H(z)| &= \sqrt{H(z)\overline{H(z)}} \\ &= \sqrt{\frac{(1 + z^{-1})(1 + z^1)}{4}} \\ &= \sqrt{\frac{2 + z^1 + z^{-1}}{4}} \\ |H_k| &= \sqrt{\frac{1 + \cos(2\pi k/N)}{2}} \end{aligned}$$

図示してみれば、まあ、低域通過フィルタ(LPF)と云えるでしょう

If we draw it graphically, we can say that it is a low-pass filter (LPF)

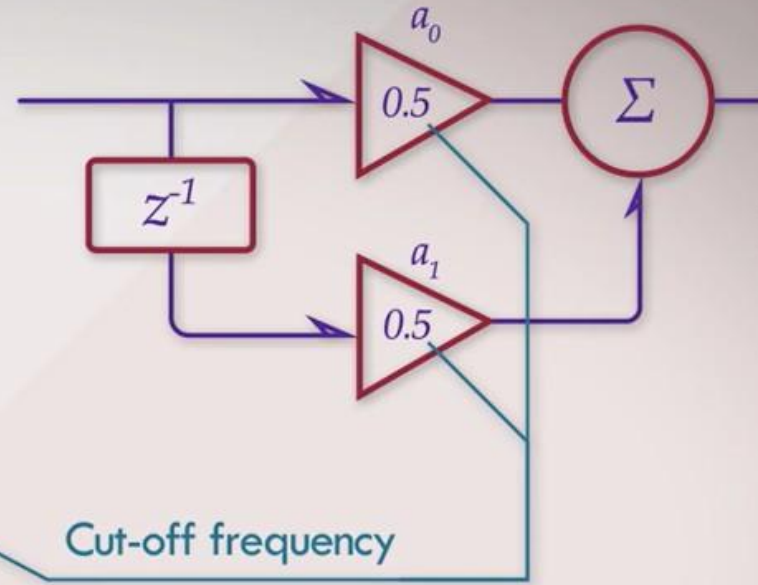
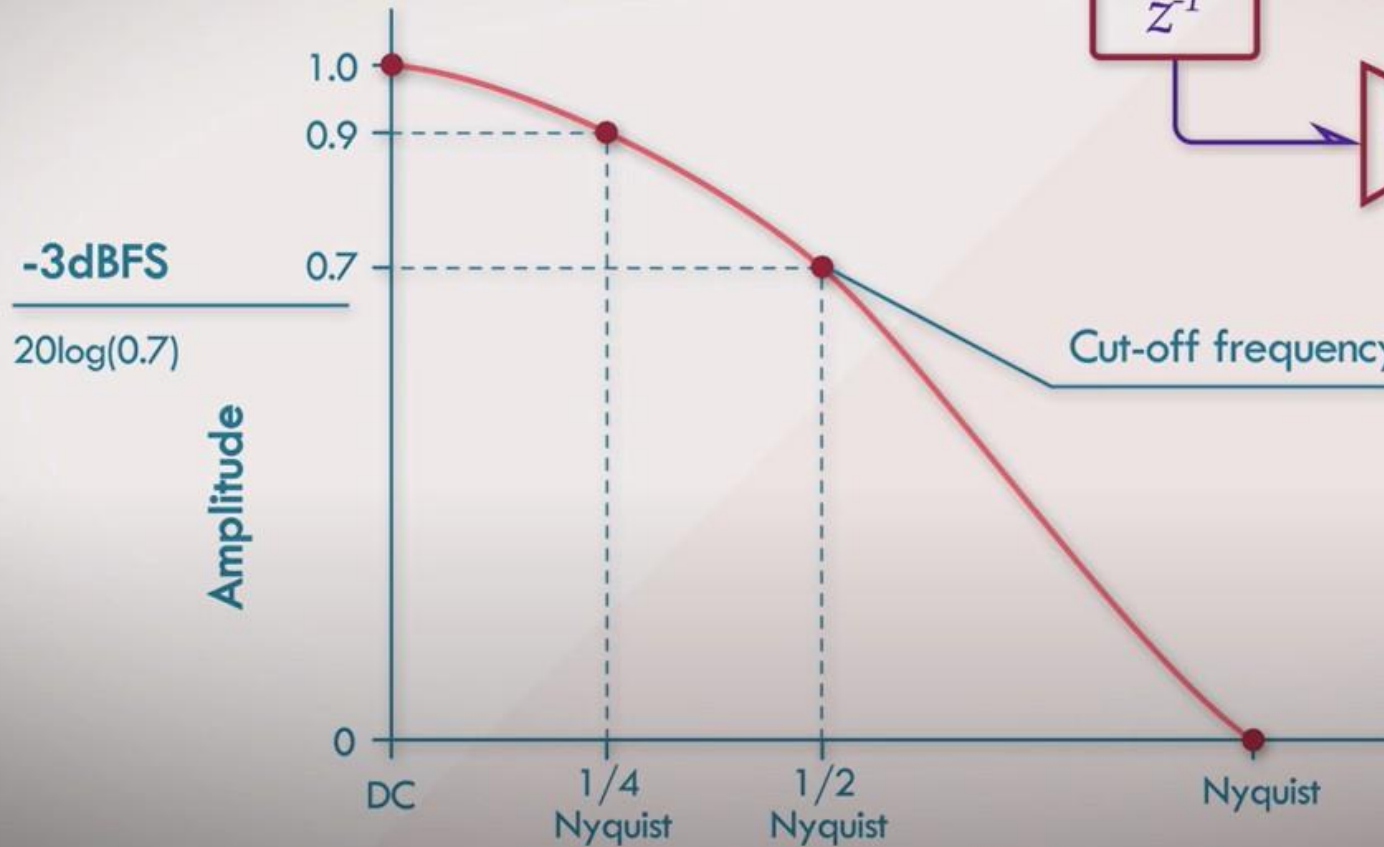
なお、グラフは周波数番号 $k=N/2$ を中心として対称形で、右半分(高域側)も極大となっているので、BEF のようにも見えますね。しかし、右半分は高域ではなく、負の周波数の折返しなので無視するんですよ

The graph is symmetrical around the frequency number $k = N/2$, and the right half (high frequency side) also has a maximum, so it looks like a BEF. However, the right half is not a high frequency, but a negative frequency wraparound, so we can ignore it.

遮断周波数は？ 伝達関数の最大値の $1/\sqrt{2}$ となる周波数番号が $k=N/4$ なので、 $f_0=f_s/4$ ということです

What is the cutoff frequency? The frequency number where the transfer function has a maximum value of $1/\sqrt{2}$ is $k=N/4$, so $f_0=f_s/4$.

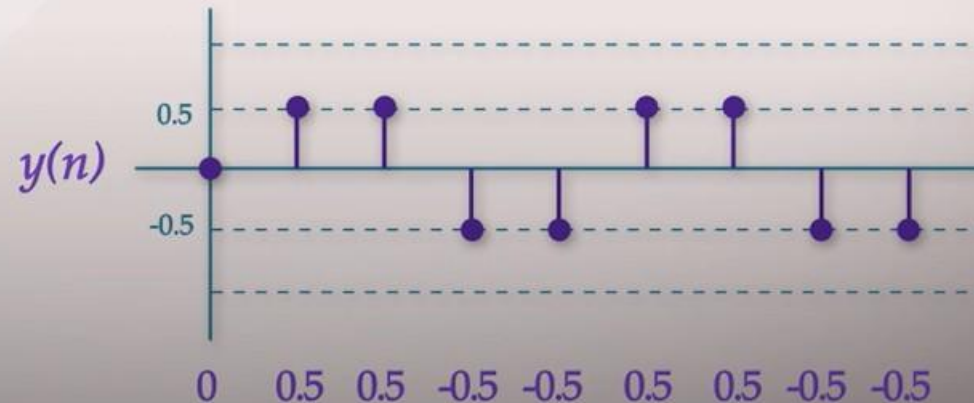
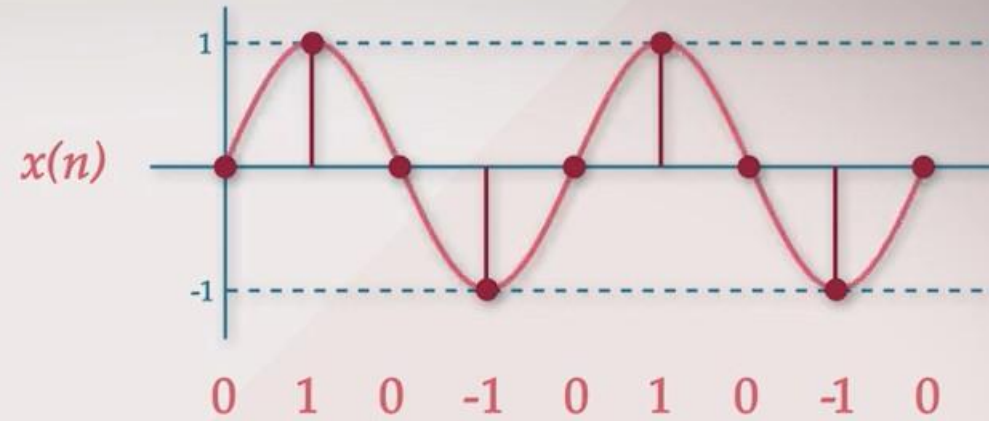
Frequency response



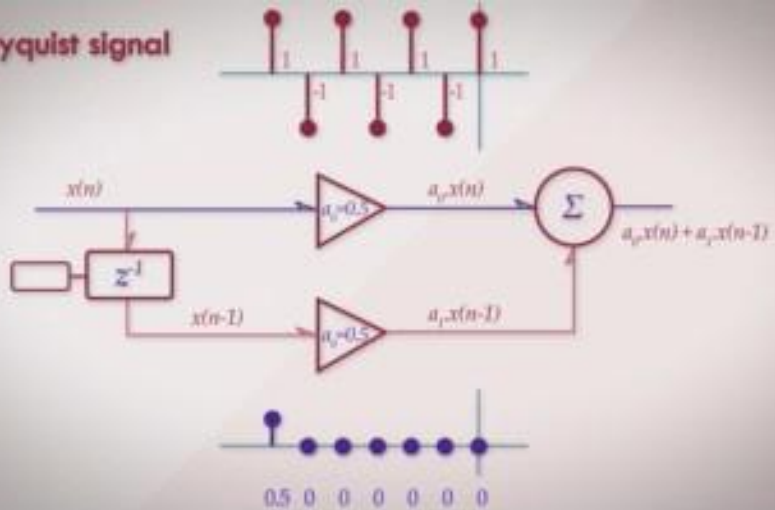


1/2 Nyquist signal

$x(n)$	$y(n)$ $0.5*x(n) + 0.5*x(n-1)$
0	0
1	0.5
0	0.5
-1	-0.5
0	-0.5
1	0.5
0	0.5
-1	-0.5
0	-0.5



Nyquist signal



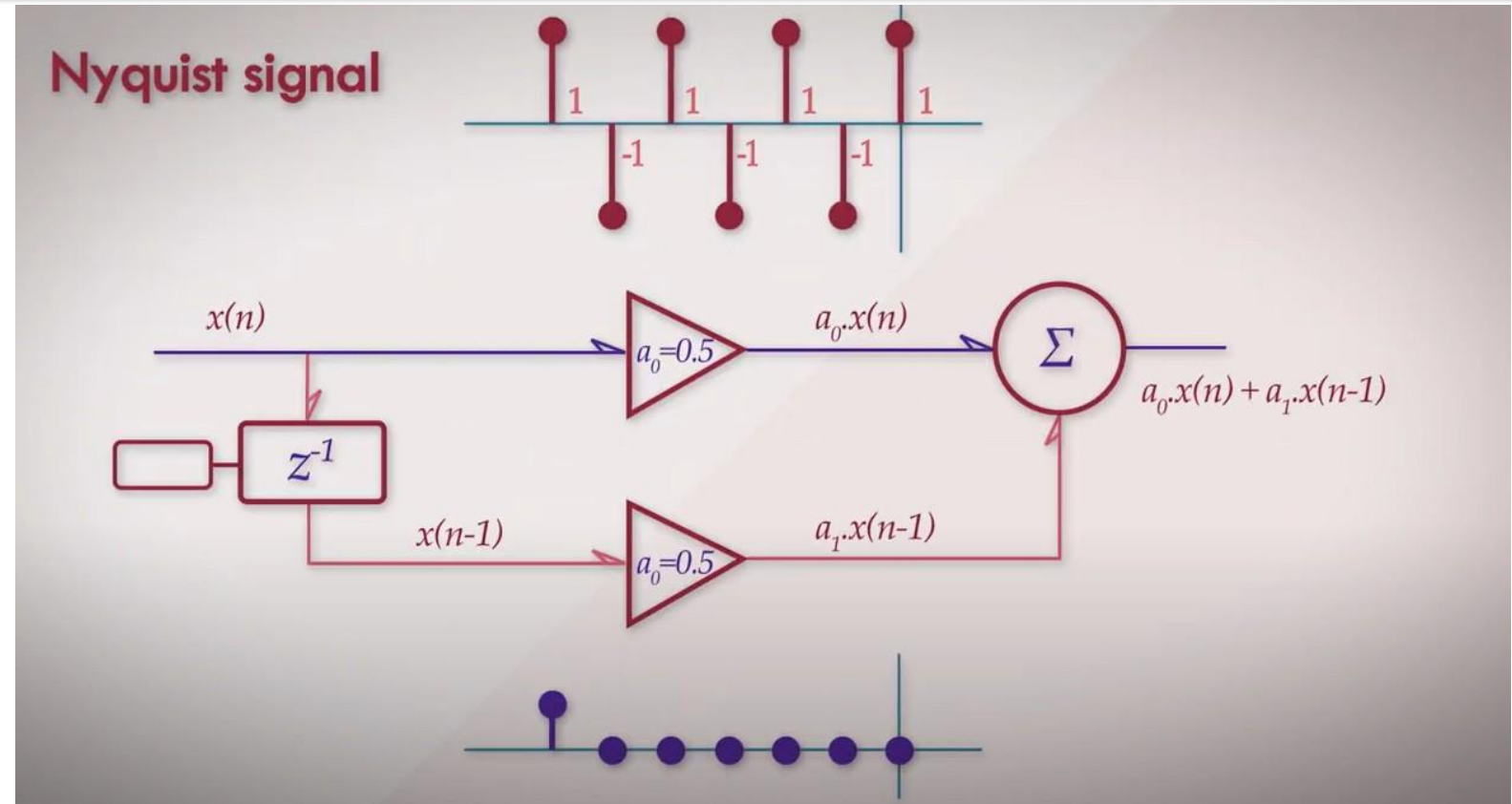
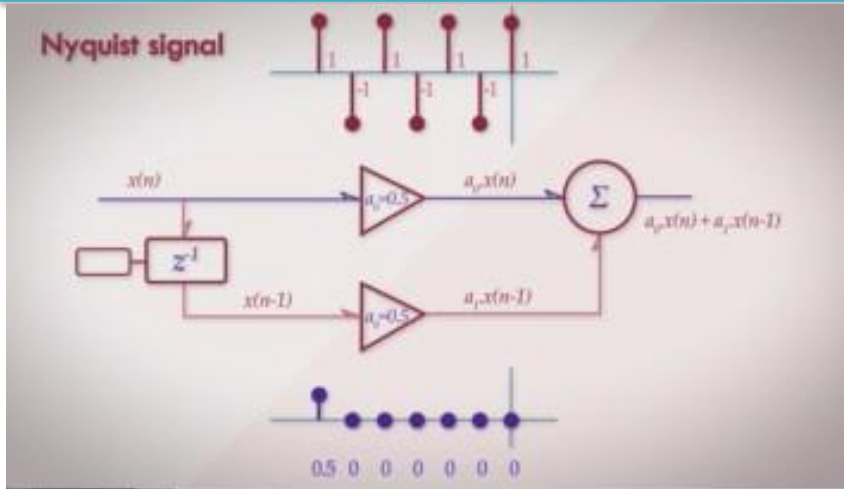
impulse signal



impulse response



練習問題





デジタル信号処理との関係 Relationship with DSP

